Double-Factorial Transseries and Their Inversion

Bernoulli–Bell Coefficients, Lambert-W Cores,
Parity Sectors, Borel Structure, and Discrete Inverses

A self-contained coefficient-level research synthesis

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Abstract

The double factorial is deceptively close to the ordinary factorial, yet its asymptotic inversion contains three structural features that must be separated: a parity sector, a power–logarithmic growth law, and a divergent Bernoulli correction series. This article gives an all-orders treatment of all three.

For the parity label $\varepsilon \in \{0, 1\}$, define

$$D_\varepsilon(x) = C_\varepsilon 2^{x/2}\Gamma\left(\frac{x}{2}+1\right), \qquad C_\varepsilon = \left(\frac{2}{\pi}\right)^{\varepsilon/2}.$$

Then $D_0(n) = n!!$ for even n, while $D_1(n) = n!!$ for odd n. The centered coordinate $u = x+1$ gives the exact normal form

$$\log D_\varepsilon(x) = \frac{u}{2}(\log u - 1) + A_\varepsilon + \Omega(u), \qquad A_0 = \frac{1}{2}\log \pi, \quad A_1 = \frac{1}{2}\log 2,$$

where the same correction Ω serves both parity sectors. We prove the Borel–Laplace representation

$$\Omega(u) = \int_0^\infty \mathrm{e}^{-us}\frac{s/\sinh s - 1}{2s^2}\, ds$$

and obtain the complete logarithmic expansion

$$\Omega(u) \sim \sum_{k\geq 1}\frac{c_k}{u^{2k-1}}, \qquad c_k = \frac{(1-2^{2k-1})B_{2k}}{2k(2k-1)} = (-1)^k\frac{(2k-2)!\eta(2k)}{\pi^{2k}}.$$

An exact signed remainder formula and a simple explicit remainder bound follow from a Mittag–Leffler expansion of the Borel kernel. Exponentiation is handled to arbitrary order by complete Bell polynomials, yielding closed partition sums and a triangular recurrence for every multiplicative coefficient.

For inversion, put

$$R_\varepsilon = \log y - A_\varepsilon, \qquad v = \mathrm{W}_0\left(\frac{2R_\varepsilon}{\mathrm{e}}\right), \qquad X = \frac{2R_\varepsilon}{v} = \mathrm{e}^{v+1}, \qquad \Lambda = v+1 = \log X.$$

The exact Lambert core solves $\frac{1}{2}X(\log X - 1) = R_\varepsilon$. The inverse has the universal expansion

$$D_\varepsilon^{-1}(y) \sim X - 1 + \sum_{n\geq 1}\frac{d_{2n-1}(\Lambda)}{X^{2n-1}},$$

where parity occurs only through R_ε. We derive: (i) a triangular coefficient-extraction recurrence; (ii) a scalar formula in ordinary partial Bell polynomials; and (iii) a nonrecursive Lagrange–Bürmann formula involving the compositional inverse of the Bernoulli series. In particular, $d_{2n-1}(\Lambda) \in \Lambda^{-1}\mathbb{Q}[\Lambda^{-1}]$, its largest possible inverse power is $\Lambda^{-(2n-1)}$, and its leading term is $-2c_n/\Lambda$. The Lambert core is then expanded into iterated logarithms with unsigned Stirling cycle numbers, and a second Bell-polynomial extraction formula gives every coefficient of the resulting polynomial–logarithmic transseries.

The article also treats the exact generalized inverse of the discrete sequence, the standard all-integer analytic interpolation and its oscillatory inverse, complex inverse sheets, optimal truncation and Borel singularities, symbolic algorithms, numerical checks, and the extension of the entire calculus to multifactorials.

Contents

Reader's guide

The main forward theorem is theorem 4.1; the multiplicative coefficient formulas are in theorem 4.3. The inverse normal form is derived in theorem 6.1; its coefficient recurrence and closed Lagrange formula are theorems 6.2 and 6.5. The complete iterated-logarithm expansion is encoded in theorem 9.1. Readers concerned with the inverse of the *integer sequence*, rather than an analytic parity branch, should also read chapter 11. The distinction between parity-resolved inverses and the inverse of a single oscillatory interpolation is explained in chapter 12.

Chapter 1

Introduction and statement of the problem

1.1 Why the word "inverse" needs a definition

For $n \in \mathbb{N}$, the double factorial is

$$n!! = n(n-2)(n-4)\cdots, \qquad 0!! = (-1)!! = 1.$$

The sequence begins

$$1, 1, 2, 3, 8, 15, 48, 105, 384, 945, \ldots$$

and is OEIS A006882 [1]. There are three different objects that are routinely called an inverse double factorial.

(i) One may invert a smooth continuation of the even subsequence.

(ii) One may invert a smooth continuation of the odd subsequence.

(iii) One may ask for the least integer n such that $n!! \geq y$.

A fourth possibility is to choose one analytic function that interpolates both parities and invert that function. Such an interpolation is not unique, and its inverse generally differs from both parity-resolved inverses by a correction larger than the first Bernoulli correction. Any complete answer must therefore state which inverse is being expanded.

This article first solves the parity-resolved problem, which is canonical for the asymptotics of the sequence. It then converts those answers into an exact formula for the discrete generalized inverse. Finally, it analyzes the all-integer interpolation recorded in OEIS and exhibits the extra oscillatory transseries generated by its periodic multiplier.

1.2 Relation to coefficient calculus and earlier inversion work

The derivation is organized around the coefficient operations emphasized in the ProveIt *Combinatorial Coefficient Calculus*: exponentiation by complete Bell polynomials, composition by partial Bell polynomials, and reversion by Lagrange–Bürmann extraction [12]. The Lambert-core strategy is also aligned with the special-function inversion program in the repository's series-and-transseries collection: invert the dominant power–logarithmic phase exactly, then generate correction blocks above that core [13, 14]. Here these ideas are specialized, proved, and sharpened for the double factorial. In particular, the centered double-factorial correction has an unusually simple Borel transform, allowing more precise remainder and late-term information than a purely formal calculation provides.

1.3 What "full transseries" means here

We distinguish three levels.

Definition 1.1 (Algebraic all-orders transseries)**.** *An expansion is algebraically complete if every inverse-power and iterated-log coefficient is specified by a finite coefficient-extraction formula or a triangular recurrence, and if fixed-order remainders are controlled.*

Definition 1.2 (Borel or hyperasymptotic completion)**.** *A Borel completion supplements the divergent algebraic series by the Borel singularities, terminant contributions, and exponentially small optimal remainders that occur beyond all algebraic orders.*

The main formulas of this article are algebraically complete in the first sense. For the forward correction we also give its exact Borel transform and pole lattice. This determines the beyond-all-orders scale and transports it through the inverse map. We do not disguise a terminant expansion as an ordinary power series: the algebraic, iterated-logarithmic, and hyperasymptotic levels are kept separate.

Main result

For both parities, all algebraic inverse coefficients are generated by one universal recurrence. The only parity dependence is the replacement

$$\log y - \tfrac{1}{2}\log\pi \quad \longleftrightarrow \quad \log y - \tfrac{1}{2}\log 2$$

inside the Lambert-W core.

Chapter 2

Parity-resolved analytic models

2.1 Gamma representations

For $m \geq 0$,

$$(2m)!! = 2^m m!, \qquad (2m+1)!! = \frac{2^{m+1}\Gamma(m+\frac{3}{2})}{\sqrt{\pi}}.$$

Equivalently, with a parity label $\varepsilon \in \{0,1\}$, define

$$D_\varepsilon(x) := C_\varepsilon 2^{x/2}\Gamma\left(\frac{x}{2}+1\right), \qquad C_\varepsilon := \left(\frac{2}{\pi}\right)^{\varepsilon/2}. \tag{2.1}$$

Then

$$D_0(2m) = (2m)!!, \qquad D_1(2m+1) = (2m+1)!!.$$

The two branch functions differ only by a constant:

$$D_1(x) = \sqrt{\frac{2}{\pi}}\, D_0(x).$$

Proposition 2.1 (Monotonicity)**.** *Each D_ε is strictly increasing on $[0,\infty)$, hence has a real inverse on its range.*

Proof. Logarithmic differentiation gives

$$\frac{D'_\varepsilon(x)}{D_\varepsilon(x)} = \frac{1}{2}\left(\log 2 + \psi\left(\frac{x}{2}+1\right)\right),$$

where $\psi = \Gamma'/\Gamma$. The trigamma function is positive on the positive real axis, so ψ is increasing. At $x = 0$, the bracket is $\log 2 - \gamma > 0$. It remains positive for every $x \geq 0$. $\square$

2.2 The centered coordinate

The ordinary large-x Stirling expansion contains a half-logarithm:

$$\log D_\varepsilon(x) \sim \frac{x}{2}(\log x - 1) + \frac{1}{2}\log x + A_\varepsilon + \cdots .$$

For inversion it is advantageous to absorb that half-logarithm into the power–logarithmic phase. Set

$$u := x+1, \qquad A_\varepsilon := \log(C_\varepsilon\sqrt{\pi}) = \begin{cases} \frac{1}{2}\log\pi, & \varepsilon = 0, \\ \frac{1}{2}\log 2, & \varepsilon = 1. \end{cases} \tag{2.2}$$

The choice $u = x + 1$ is not cosmetic: it converts the gamma argument into $(u+1)/2 = u/2 + 1/2$, and Bernoulli polynomials at $1/2$ kill every odd correction.

Theorem 2.2 (Exact centered normal form). *For $x > -1$, $u = x + 1$, and $\varepsilon \in \{0, 1\}$,*

$$D_\varepsilon(x) = \exp(A_\varepsilon) \left(\frac{u}{\mathrm{e}}\right)^{u/2} \exp\bigl(\Omega(u)\bigr), \tag{2.3}$$

where

$$\Omega(u) := \log\Gamma\left(\frac{u+1}{2}\right) - \frac{u}{2}\log\left(\frac{u}{2}\right) + \frac{u}{2} - \frac{1}{2}\log(2\pi). \tag{2.4}$$

In particular, Ω is independent of parity.

Proof. Insert $x = u - 1$ in equation (2.1):

$$\log D_\varepsilon(x) = \log C_\varepsilon + \frac{u-1}{2}\log 2 + \log\Gamma\left(\frac{u+1}{2}\right).$$

Subtract $A_\varepsilon + \frac{u}{2}(\log u - 1)$, use $A_\varepsilon = \log C_\varepsilon + \frac{1}{2}\log\pi$, and simplify. The remainder is exactly $\Omega(u)$ in equation (2.4). $\square$

2.3 A single parity transmonomial

For integer n, the constants in equation (2.2) can be encoded by

$$A(n) = \frac{1}{4}\log(2\pi) + \frac{(-1)^n}{4}\log\left(\frac{\pi}{2}\right).$$

Consequently the exact sequence-level factorization is

$$n!! = \left(\frac{n+1}{\mathrm{e}}\right)^{(n+1)/2} \exp\left\{\frac{1}{4}\log(2\pi) + \frac{(-1)^n}{4}\log\left(\frac{\pi}{2}\right) + \Omega(n+1)\right\}. \tag{2.5}$$

Thus $(-1)^n$ is a genuine parity transmonomial. The algebraic correction $\Omega(n+1)$ is common to both sectors.

Chapter 3

Exact Borel structure of the centered correction

3.1 A Binet-type integral

Theorem 3.1 (Borel–Laplace representation)**.** *For* $\Re u > 0$*,*

$$\Omega(u) = \int_0^\infty \mathrm{e}^{-us}\, \widehat{\Omega}(s)\, ds, \qquad \widehat{\Omega}(s) := \frac{1}{2s^2}\left(\frac{s}{\sinh s} - 1\right). \tag{3.1}$$

The apparent singularity of $\widehat{\Omega}$ *at* $s = 0$ *is removable.*

Proof. It is convenient to put $t = u/2$ and write

$$\widetilde{\Omega}(t) := \log\Gamma(t + \tfrac{1}{2}) - t\log t + t - \tfrac{1}{2}\log(2\pi).$$

The standard integral representation of the digamma function, combined with Frullani's formula for $\log t$, gives

$$\psi(t + \tfrac{1}{2}) - \log t = \int_0^\infty \mathrm{e}^{-tr}\left(\frac{1}{r} - \frac{1}{2\sinh(r/2)}\right) dr.$$

On the other hand,

$$\frac{d}{dt}\int_0^\infty \mathrm{e}^{-tr}\,\frac{r/(2\sinh(r/2)) - 1}{r^2}\, dr = \int_0^\infty \mathrm{e}^{-tr}\left(\frac{1}{r} - \frac{1}{2\sinh(r/2)}\right) dr.$$

Both this integral and $\widetilde{\Omega}(t)$ tend to zero as $t \to +\infty$; hence they are equal. Finally substitute $r = 2s$. The Taylor expansion $s/\sinh s = 1 - s^2/6 + \mathrm{O}(s^4)$ removes the origin. $\square$

3.2 Mittag–Leffler decomposition

The elementary Mittag–Leffler expansion

$$\frac{s}{\sinh s} = 1 + 2s^2 \sum_{m=1}^{\infty} \frac{(-1)^m}{s^2 + \pi^2 m^2} \tag{3.2}$$

immediately gives

$$\widehat{\Omega}(s) = \sum_{m=1}^{\infty} \frac{(-1)^m}{s^2 + \pi^2 m^2}. \tag{3.3}$$

The Borel singularities are therefore the simple poles

$$s = \pm \mathrm{i}\pi m, \qquad m = 1, 2, \ldots.$$

Their residues are

$$\operatorname*{Res}_{s=\mathrm{i}\pi m} \widehat{\Omega}(s) = \frac{(-1)^{m+1}\mathrm{i}}{2\pi m}, \tag{3.4}$$

$$\operatorname*{Res}_{s=-\mathrm{i}\pi m} \widehat{\Omega}(s) = \frac{(-1)^{m}\mathrm{i}}{2\pi m}. \tag{3.5}$$

The nearest pair, at distance π from the origin, controls the factorial divergence and the optimal remainder scale.

Chapter 4

The forward double-factorial transseries

4.1 The logarithmic coefficients

Let B_n denote the Bernoulli numbers, with $B_1 = -1/2$, and let

$$\eta(s) = \sum_{m=1}^{\infty} \frac{(-1)^{m-1}}{m^s} = (1 - 2^{1-s})\zeta(s)$$

be the Dirichlet eta function.

Theorem 4.1 (All-orders logarithmic expansion). *As $u \to \infty$ in any closed subsector of $|\arg u| < \pi/2$,*

$$\Omega(u) \sim \sum_{k=1}^{\infty} \frac{c_k}{u^{2k-1}}, \tag{4.1}$$

where the following three formulas are equivalent:

$$\boxed{c_k = \frac{2^{2k-1} B_{2k}(\frac{1}{2})}{2k(2k-1)} = \frac{(1-2^{2k-1})B_{2k}}{2k(2k-1)} = (-1)^k \frac{(2k-2)!\eta(2k)}{\pi^{2k}}.} \tag{4.2}$$

For real $u > 0$, truncation after N terms has the exact remainder

$$R_N(u) = (-1)^N \int_0^{\infty} \mathrm{e}^{-us} s^{2N} \sum_{m=1}^{\infty} \frac{(-1)^m}{(\pi m)^{2N}(s^2 + \pi^2 m^2)}\, ds, \tag{4.3}$$

and therefore

$$0 < (-1)^{N+1} R_N(u) < \frac{(2N)!}{\pi^{2N+2} u^{2N+1}}. \tag{4.4}$$

Proof. Expand each summand of equation (3.3):

$$\frac{1}{s^2 + a^2} = \sum_{j=0}^{N-1} \frac{(-1)^j s^{2j}}{a^{2j+2}} + \frac{(-1)^N s^{2N}}{a^{2N}(s^2 + a^2)}.$$

Termwise Laplace integration of the finite sum gives

$$c_{j+1} = (2j)! \frac{(-1)^{j+1}\eta(2j+2)}{\pi^{2j+2}},$$

which is the last expression in equation (4.2). Euler's formula for B_{2k} and $B_{2k}(1/2) = (2^{1-2k} - 1)B_{2k}$ give the other two forms [4]. The remaining geometric-series term is equation (4.3).

For fixed $s > 0$, the positive sequence

$$\frac{1}{m^{2N}(s^2 + \pi^2 m^2)}$$

strictly decreases with m. Its alternating sum, beginning with a negative term, is negative and has magnitude smaller than its first term. This proves the sign assertion. Finally,

$$|R_N(u)| < \int_0^\infty \frac{\mathrm{e}^{-us} s^{2N}}{\pi^{2N}(s^2 + \pi^2)}\, ds < \frac{1}{\pi^{2N+2}} \int_0^\infty \mathrm{e}^{-us} s^{2N}\, ds,$$

which is equation (4.4). □

The first coefficients are

$$\Omega(u) \sim -\frac{1}{12u} + \frac{7}{360u^3} - \frac{31}{1260u^5} + \frac{127}{1680u^7} - \frac{511}{1188u^9} + \frac{1414477}{360360u^{11}} - \cdots. \tag{4.5}$$

Corollary 4.2 (Forward logarithmic transseries). *For $x \to +\infty$, $u = x + 1$, and fixed ε,*

$$\boxed{\log D_\varepsilon(x) \sim \frac{u}{2}(\log u - 1) + A_\varepsilon + \sum_{k=1}^{\infty} \frac{c_k}{u^{2k-1}}.} \tag{4.6}$$

For integer n, replace A_ε by $A(n)$ from equation (2.5).

4.2 Late coefficients and optimal truncation

From equation (4.2),

$$c_k = (-1)^k \frac{(2k-2)!}{\pi^{2k}} \left(1 + \mathrm{O}(2^{-2k})\right). \tag{4.7}$$

Thus the logarithmic series is Gevrey−1 when indexed by its actual power. The ratio of consecutive term magnitudes is asymptotic to

$$\frac{(2k)(2k-1)}{\pi^2 u^2}.$$

The least term occurs near $2k \approx \pi u$, and its size is

$$\asymp u^{-1/2} \mathrm{e}^{-\pi u}. \tag{4.8}$$

This agrees with the distance π to the nearest Borel poles. On the positive real axis the series alternates and is Borel summable without a lateral ambiguity. In complex directions, the poles $\pm \mathrm{i}\pi m$ generate the usual terminant and Stokes structure for shifted gamma asymptotics; see [9, 10, 11].

4.3 Multiplicative coefficients by Bell polynomials

Define

$$\gamma_j := \begin{cases} c_{(j+1)/2}, & j \text{ odd}, \\ 0, & j \text{ even}. \end{cases} \tag{4.9}$$

Then

$$\exp\left(\sum_{k \geq 1} c_k u^{-(2k-1)}\right) = \sum_{m \geq 0} \frac{s_m}{u^m}.$$

Theorem 4.3 (Complete Bell-polynomial formula)**.** *The multiplicative coefficients are given by any of the equivalent formulas*

$$s_m = \frac{1}{m!} Y_m(1!\gamma_1, 2!\gamma_2, \ldots, m!\gamma_m), \tag{4.10}$$

$$= \sum_{\substack{k_1,\ldots,k_m \geq 0 \\ k_1+2k_2+\cdots+mk_m=m}} \prod_{j=1}^{m} \frac{\gamma_j^{k_j}}{k_j!}, \tag{4.11}$$

$$s_0 = 1, \qquad m s_m = \sum_{j=1}^{m} j\gamma_j s_{m-j} \quad (m \geq 1). \tag{4.12}$$

Consequently,

$$\boxed{D_\varepsilon(x) \sim \mathrm{e}^{A_\varepsilon} \left(\frac{u}{\mathrm{e}}\right)^{u/2} \sum_{m=0}^{\infty} \frac{s_m}{u^m}.} \tag{4.13}$$

Proof. Put $t = u^{-1}$ and $S(t) = \exp(\sum_{j\geq1} \gamma_j t^j)$. The exponential formula for complete Bell polynomials gives equation (4.10); direct multinomial expansion gives equation (4.11). Differentiating $S(t)$ and comparing coefficients in

$$tS'(t) = \left(\sum_{j\geq1} j\gamma_j t^j\right) S(t)$$

gives equation (4.12). □

The first coefficients are displayed in table 4.1.

Table 4.1: First multiplicative coefficients in equation (4.13).

m	s_m
0	1
1	$-1/12$
2	$1/288$
3	$1003/51840$
4	$-4027/2488320$
5	$-5128423/209018880$
6	$168359651/75246796800$
7	$68168266699/902961561600$
8	$-587283555451/86684309913600$

4.4 The uncentered form

For comparison with conventional Stirling formulas, direct expansion at $x = \infty$ gives

$$\log D_\varepsilon(x) \sim \frac{x}{2}(\log x - 1) + \frac{1}{2}\log x + A_\varepsilon + \sum_{k=1}^{\infty} \frac{b_k}{x^{2k-1}}, \tag{4.14}$$

where

$$b_k = \frac{2^{2k-1} B_{2k}}{2k(2k-1)}. \tag{4.15}$$

The leading term is therefore

$$D_\varepsilon(x) \sim \mathrm{e}^{A_\varepsilon} \sqrt{x} \left(\frac{x}{\mathrm{e}}\right)^{x/2},$$

which yields the familiar constants $\sqrt{\pi}$ for even n and $\sqrt{2}$ for odd n [1].

The centered and uncentered coefficients satisfy the identity

$$\frac{(-1)^{r+1}}{2r(r+1)} + (-1)^{r+1} \sum_{k=1}^{\lfloor (r+1)/2 \rfloor} \binom{r-1}{2k-2} c_k = \begin{cases} b_{(r+1)/2}, & r \text{ odd}, \\ 0, & r \text{ even}. \end{cases} \tag{4.16}$$

It follows simply by substituting $u = x+1$ into equation (4.6) and comparing powers. The cancellation of every even power is another way to see why the shift $u = x + 1$ is natural.

Chapter 5

The exact Lambert core

5.1 Dominant phase inversion

Fix $\varepsilon \in \{0,1\}$ and let $y \to +\infty$. Set

$$R_\varepsilon := \log y - A_\varepsilon. \tag{5.1}$$

Ignoring $\Omega(u)$ in equation (2.3) leaves

$$\frac{u}{2}(\log u - 1) = R_\varepsilon. \tag{5.2}$$

Proposition 5.1 (Lambert solution of the core)**.** *Define*

$$v := \mathrm{W}_0\left(\frac{2R_\varepsilon}{\mathrm{e}}\right), \qquad X := \frac{2R_\varepsilon}{v} = \mathrm{e}^{v+1}, \qquad \Lambda := v + 1 = \log X. \tag{5.3}$$

Then X is the unique positive solution of equation (5.2), and

$$\frac{d}{du}\left[\frac{u}{2}(\log u - 1)\right]_{u=X} = \frac{\Lambda}{2}.$$

Proof. Writing $v = \log u - 1$ gives $u = \mathrm{e}^{v+1}$, and equation (5.2) becomes

$$v\mathrm{e}^v = \frac{2R_\varepsilon}{\mathrm{e}}.$$

For positive R_ε the principal branch W_0 is real and positive. The derivative formula is immediate. □

The exact Lambert core avoids a premature expansion in nested logarithms. This is both algebraically cleaner and numerically more stable. Iterated logarithms will be introduced only after the inverse-power correction blocks have been generated.

5.2 Scale hierarchy

As $y \to \infty$,

$$R_\varepsilon \sim \log y, \qquad v \sim \log R_\varepsilon, \qquad X \sim \frac{2R_\varepsilon}{\log R_\varepsilon}, \qquad \Lambda \sim \log R_\varepsilon.$$

The first forward correction is $\Omega(X) \sim -1/(12X)$. Division by the core slope $\Lambda/2$ shows that the first displacement of the inverse is of size

$$\frac{1}{6\Lambda X} \sim \frac{1}{12R_\varepsilon}.$$

This scale is invisible if one retains only the leading Lambert core.

Chapter 6

Universal all-orders inversion

6.1 The normalized implicit equation

Write the exact solution in relative form

$$u = X(1+E), \qquad q := X^{-1}. \tag{6.1}$$

Define

$$H(E) := (1+E)\log(1+E) - E = \sum_{m=2}^{\infty} \frac{(-1)^m}{m(m-1)} E^m \tag{6.2}$$

and

$$a_k := 2c_k = \frac{(1-2^{2k-1})B_{2k}}{k(2k-1)}. \tag{6.3}$$

Theorem 6.1 (Master inverse equation). *After formal substitution of the complete asymptotic series for Ω, the inverse equation is*

$$\boxed{\Lambda E + H(E) + \sum_{k=1}^{\infty} a_k q^{2k}(1+E)^{-(2k-1)} = 0.} \tag{6.4}$$

It has a unique formal solution of the form

$$E(q;\Lambda) = \sum_{n=1}^{\infty} e_n(\Lambda) q^{2n}. \tag{6.5}$$

Therefore

$$\boxed{D_\varepsilon^{-1}(y) \sim X - 1 + \sum_{n=1}^{\infty} \frac{d_{2n-1}(\Lambda)}{X^{2n-1}}, \qquad d_{2n-1} = e_n.} \tag{6.6}$$

No even inverse powers of X occur.

Proof. Because X solves the core equation,

$$\begin{aligned}\frac{X(1+E)}{2}\big(\log(X(1+E)) - 1\big) - \frac{X}{2}(\log X - 1)\\ = \frac{X}{2}\left[\Lambda E + (1+E)\log(1+E) - E\right] = \frac{X}{2}\big(\Lambda E + H(E)\big).\end{aligned}$$

The exact equation is the core difference plus $\Omega(X(1+E)) = 0$. Multiplying by $2/X = 2q$, then substituting

$$\Omega(X(1+E)) \sim \sum_{k\geq 1} c_k q^{2k-1}(1+E)^{-(2k-1)}$$

gives equation (6.4). Its linear term is ΛE, with $\Lambda \neq 0$. Formal implicit inversion therefore gives a unique solution. Since every forcing term begins with an even power of q, and H has no constant or linear term, induction shows that E contains only q^{2n}. Finally, $u - X = XE = \sum e_n X^{1-2n}$, and $x = u - 1$. □

6.2 Triangular coefficient extraction

Let

$$E_{n-1}(q) := \sum_{j=1}^{n-1} e_j(\Lambda) q^{2j}.$$

Theorem 6.2 (Universal triangular recurrence). *For $n \geq 1$,*

$$\boxed{e_n(\Lambda) = -\frac{1}{\Lambda}\left[q^{2n}\right] H(E_{n-1}) + \sum_{k=1}^{n} a_k q^{2k}(1+E_{n-1})^{-(2k-1)}.} \tag{6.7}$$

Every coefficient on the right is already known when e_n is computed.

Proof. Insert $E = E_{n-1} + e_n q^{2n} + \mathrm{O}(q^{2n+2})$ into equation (6.4). The term ΛE contributes $\Lambda e_n q^{2n}$. Since $H(E) = \mathrm{O}(E^2)$, no occurrence of $e_n q^{2n}$ can contribute to the coefficient of q^{2n} in $H(E)$. In the k-th forcing term the required residual degree is $2(n-k) < 2n$, so e_n again cannot occur. Solving the coefficient equation gives equation (6.7). □

Coefficient mechanism

The recurrence is finite at every order. To obtain e_n, one needs only $a_1, \ldots, a_n$ and $e_1, \ldots, e_{n-1}$. No asymptotic guessing, numerical fitting, or re-expansion of Lambert W is required.

6.3 Scalar Bell-polynomial formula

For a series $E(z) = \sum_{j\geq 1} e_j z^j$, define the ordinary partial Bell convolution

$$\widehat{B}_{r,m}(e_1, e_2, \ldots) := [z^r]\, E(z)^m = \frac{m!}{r!} B_{r,m}(1!e_1, 2!e_2, \ldots), \tag{6.8}$$

with

$$\widehat{B}_{0,0} = 1, \qquad \widehat{B}_{r,0} = 0\ (r > 0).$$

Corollary 6.3 (Bell-polynomial recurrence). *The coefficients in equation (6.5) satisfy*

$$e_n = -\frac{1}{\Lambda}\left[\sum_{m=2}^{n} \frac{(-1)^m}{m(m-1)} \widehat{B}_{n,m} + \sum_{k=1}^{n} a_k \sum_{m=0}^{n-k} (-1)^m \binom{2k+m-2}{m} \widehat{B}_{n-k,m}\right]. \tag{6.9}$$

All Bell polynomials in this formula involve only $e_1, \ldots, e_{n-1}$.

Proof. Use the series for H in equation (6.2) and

$$(1+E)^{-(2k-1)} = \sum_{m\geq 0}(-1)^m\binom{2k+m-2}{m}E^m.$$

Extract the coefficient of z^n, where $z=q^2$, and apply equation (6.8). □

6.4 Denominator and leading-term structure

Theorem 6.4 (Polynomial structure). *For every $n\geq 1$,*

$$e_n(\Lambda)\in\Lambda^{-1}\mathbb{Q}[\Lambda^{-1}], \qquad \deg_{\Lambda^{-1}} e_n\leq 2n-1. \tag{6.10}$$

More precisely,

$$e_n(\Lambda) = -\frac{a_n}{\Lambda}+\mathrm{O}(\Lambda^{-2}) \qquad (\Lambda\to\infty). \tag{6.11}$$

Equivalently,

$$d_{2n-1}(\Lambda) = -\frac{2c_n}{\Lambda}+\mathrm{O}(\Lambda^{-2}).$$

Proof. The assertion is immediate for $n=1$. Assume it for all smaller indices. Every monomial in $\widehat{B}_{r,m}$ is a product of m earlier coefficients whose indices sum to r. Its least power of Λ^{-1} is m, while its largest is at most

$$\sum_{\nu=1}^{m}(2j_\nu-1) = 2r-m.$$

The outer factor Λ^{-1} in equation (6.9) raises the largest possible exponent by one. In the first sum, $r=n$ and $m\geq 2$, giving at most $2n-1$. In the forcing sum, $r=n-k$, and multiplication by a_k does not affect Λ; the same bound follows. Rationality is preserved by every operation.

For the leading term as $\Lambda\to\infty$, all Bell terms contain at least one earlier $e_j=\mathrm{O}(\Lambda^{-1})$. The only term of order Λ^{-1} after the outer division is the $k=n, m=0$ forcing term, namely $-a_n/\Lambda$. □

6.5 A nonrecursive Lagrange–Bürmann formula

Define the Bernoulli forcing series

$$A(z) := \sum_{k=1}^{\infty} a_k z^k, \qquad a_1=-\frac{1}{6}, \tag{6.12}$$

and let $S=A^{\langle -1\rangle}$ be its formal compositional inverse.

Equation equation (6.4), with $z=q^2$, can be rewritten as

$$\Lambda E + H(E) + (1+E)A\left(\frac{z}{(1+E)^2}\right) = 0. \tag{6.13}$$

Consequently,

$$z = G_\Lambda(E) := (1+E)^2 S\left(-\frac{\Lambda E + H(E)}{1+E}\right). \tag{6.14}$$

Since $G'_\Lambda(0) = -\Lambda/a_1 = 6\Lambda\neq 0$, this series is revertible.

Theorem 6.5 (Closed coefficient extraction)**.** *For every $n \geq 1$,*

$$\boxed{e_n(\Lambda) = \frac{1}{n}\left[E^{n-1}\right]\left(\frac{E}{G_\Lambda(E)}\right)^n.} \tag{6.15}$$

The auxiliary inverse S itself has coefficients

$$[w^m]\,S(w) = \frac{1}{m}\left[t^{m-1}\right]\left(\frac{t}{A(t)}\right)^m. \tag{6.16}$$

Thus equations (6.15) and (6.16) form a nonrecursive finite coefficient formula for every e_n.

Proof. Apply the formal Lagrange–Bürmann theorem first to $z = G_\Lambda(E)$ and then to $w = A(t)$. Although A is a divergent Gevrey series, it is a valid formal power series with nonzero linear coefficient. Every requested finite jet uses only finitely many a_k, so no analytic convergence assumption is needed. ∎

Chapter 7

Explicit inverse blocks

7.1 The first five universal coefficients

Repeated use of equation (6.7) gives

$$d_1(\Lambda) = \frac{1}{6\Lambda}, \tag{7.1}$$

$$d_3(\Lambda) = -\frac{14\Lambda^2 + 10\Lambda + 5}{360\Lambda^3}, \tag{7.2}$$

$$d_5(\Lambda) = \frac{2232\Lambda^4 + 1176\Lambda^3 + 714\Lambda^2 + 350\Lambda + 105}{45360\Lambda^5}, \tag{7.3}$$

$$\begin{aligned} d_7(\Lambda) = -\frac{1}{5443200\Lambda^7}\Big(&822960\Lambda^6 + 292536\Lambda^5 + 136956\Lambda^4 \\ &+ 68040\Lambda^3 + 32970\Lambda^2 + 12250\Lambda + 2625\Big), \end{aligned} \tag{7.4}$$

$$\begin{aligned} d_9(\Lambda) = \frac{1}{359251200\Lambda^9}\Big(&309052800\Lambda^8 + 77920128\Lambda^7 + 26024592\Lambda^6 \\ &+ 10019592\Lambda^5 + 4516028\Lambda^4 + 2023560\Lambda^3 \\ &+ 812350\Lambda^2 + 242550\Lambda + 40425\Big). \end{aligned} \tag{7.5}$$

Combining these blocks gives the compact expansion through X^{-9}:

$$D_\varepsilon^{-1}(y) \sim X - 1 + \frac{d_1(\Lambda)}{X} + \frac{d_3(\Lambda)}{X^3} + \frac{d_5(\Lambda)}{X^5} + \frac{d_7(\Lambda)}{X^7} + \frac{d_9(\Lambda)}{X^9} + \cdots . \tag{7.6}$$

7.2 A compact Lambert-W version

Because $X = 2R_\varepsilon/v$ and $\Lambda = v + 1$, the first terms can be written without X:

$$\begin{aligned} D_\varepsilon^{-1}(y) \sim \frac{2R_\varepsilon}{v} - 1 + &\frac{v}{12R_\varepsilon(v+1)} \\ &- \frac{v^3\left(14(v+1)^2 + 10(v+1) + 5\right)}{2880R_\varepsilon^3(v+1)^3} \\ &+ \frac{v^5\left(2232(v+1)^4 + 1176(v+1)^3 + 714(v+1)^2 + 350(v+1) + 105\right)}{1451520R_\varepsilon^5(v+1)^5} + \cdots , \end{aligned} \tag{7.7}$$

where $v = \mathrm{W}_0(2R_\varepsilon/\mathrm{e})$. Keeping v unexpanded is generally the best numerical form.

7.3 A correction to a common first guess

A frequently used rough inversion is

$$x \approx \frac{2R_\varepsilon}{\mathrm{W}_0(2R_\varepsilon/\mathrm{e})} - 1.$$

The leading correction is not of order $1/X$ alone but

$$\frac{1}{6\Lambda X} = \frac{v}{12R_\varepsilon(v+1)} = \frac{1}{12R_\varepsilon}\left(1 - \frac{1}{v+1}\right).$$

This distinction matters when numerical inversion is desired: the factor Λ^{-1} is forced by the derivative of the power–logarithmic phase.

Chapter 8

Fixed-order error control

8.1 Residual-to-root conversion

Let

$$u_N := X + \sum_{n=1}^{N} e_n(\Lambda) X^{1-2n}, \qquad x_N := u_N - 1.$$

Theorem 8.1 (Poincaré validity of the inverse expansion)**.** *For every fixed $N \geq 0$,*

$$D_\varepsilon^{-1}(y) = X - 1 + \sum_{n=1}^{N} e_n(\Lambda) X^{1-2n} + \mathrm{O}\left(\frac{X^{-(2N+1)}}{\Lambda}\right) \tag{8.1}$$

as $y \to +\infty$. The implied constant can be chosen uniformly for $\varepsilon = 0, 1$.

Proof. By construction, substitution of $E_N = (u_N - X)/X$ into the normalized equation cancels every term through q^{2N}. The first uncancelled term is $\mathrm{O}(q^{2N+2})$, with coefficients that are rational functions of Λ bounded for large Λ. Undoing the normalization multiplies the phase residual by $X/2$, hence the residual in $\log D_\varepsilon$ is $\mathrm{O}(X^{-(2N+1)})$.

For u between u_N and the true root,

$$\frac{d}{du}\left[\frac{u}{2}(\log u - 1) + \Omega(u)\right] = \frac{1}{2}\log u + \Omega'(u) = \frac{1}{2}\Lambda\,(1 + \mathrm{o}(1)).$$

The mean-value theorem therefore divides the phase residual by a quantity asymptotic to $\Lambda/2$, proving equation (8.1). The parity constant disappears upon differentiation, so the estimate is uniform in ε. □

8.2 Backward error as a certification device

For numerical work, a truncated approximation $\widetilde{x}$ should be certified by its logarithmic residual

$$\rho_\varepsilon(\widetilde{x}; y) := \log D_\varepsilon(\widetilde{x}) - \log y.$$

Since the branch is increasing,

$$|\widetilde{x} - D_\varepsilon^{-1}(y)| \leq \frac{2|\rho_\varepsilon(\widetilde{x}; y)|}{\inf_\xi\{\log 2 + \psi(\xi/2 + 1)\}}, \tag{8.2}$$

where the infimum is over the interval between the approximation and the root. At large arguments the denominator is $\sim \Lambda$, matching equation (8.1). This backward-error test is more reliable than estimating the forward error from the last retained term alone.

Chapter 9

The complete iterated-logarithm expansion

9.1 Lambert W in Stirling-cycle form

Let

$$Z := \frac{2R_\varepsilon}{\mathrm{e}}, \qquad L_1 := \log Z, \qquad L_2 := \log L_1. \tag{9.1}$$

The large-Z expansion of the principal Lambert function can be written using unsigned Stirling cycle numbers $\left[{n \atop k}\right]$:

$$\mathrm{W}_0(Z) \sim L_1 - L_2 + \sum_{n=1}^{\infty} \frac{1}{L_1^n} \sum_{m=1}^{n} (-1)^{n+m} \left[{n \atop n-m+1}\right] \frac{L_2^m}{m!}. \tag{9.2}$$

This is the standard Stirling-number form of the Lambert expansion [6, 5].

Write

$$\frac{\mathrm{W}_0(Z)}{L_1} = 1 + \sum_{j=1}^{\infty} \frac{\alpha_j(L_2)}{L_1^j}. \tag{9.3}$$

Then

$$\alpha_1(L_2) = -L_2, \tag{9.4}$$

$$\alpha_j(L_2) = \sum_{m=1}^{j-1} (-1)^{j-1+m} \left[{j-1 \atop j-m}\right] \frac{L_2^m}{m!}, \qquad j \geq 2. \tag{9.5}$$

9.2 Reciprocal coefficients

Define $\beta_0 = 1$ and

$$\beta_n(L_2) = -\sum_{j=1}^{n} \alpha_j(L_2)\beta_{n-j}(L_2). \tag{9.6}$$

Then

$$X \sim \frac{2R_\varepsilon}{L_1} \sum_{n=0}^{\infty} \frac{\beta_n(L_2)}{L_1^n}. \tag{9.7}$$

The first polynomials are

$$\beta_0 = 1,$$
$$\beta_1 = L_2,$$

$$\beta_2 = L_2^2 - L_2,$$
$$\beta_3 = L_2^3 - \frac{5}{2}L_2^2 + L_2,$$
$$\beta_4 = L_2^4 - \frac{13}{3}L_2^3 + \frac{9}{2}L_2^2 - L_2,$$
$$\beta_5 = L_2^5 - \frac{77}{12}L_2^4 + \frac{71}{6}L_2^3 - 7L_2^2 + L_2.$$

9.3 All coefficients in the two-level transseries

Introduce $t = L_1^{-1}$ and the formal polynomial-logarithmic series

$$\mathcal{W}(t; L_2) := 1 + \sum_{j\geq 1} \alpha_j(L_2)t^j, \qquad \mathcal{L}(t; L_2) := \mathcal{W}(t; L_2) + t. \tag{9.8}$$

Then

$$\frac{W}{L_1} = \mathcal{W}, \qquad \frac{\Lambda}{L_1} = \mathcal{L}, \qquad X = \frac{2R_\varepsilon}{L_1}\mathcal{W}^{-1}.$$

Write

$$e_n(\Lambda) = \sum_{p=1}^{2n-1} e_{n,p}\Lambda^{-p}, \qquad e_{n,p} \in \mathbb{Q}. \tag{9.9}$$

Theorem 9.1 (Complete outer coefficient formula)**.** *The full algebraic inverse transseries is*

$$D_\varepsilon^{-1}(y) \sim \frac{2R_\varepsilon}{L_1}\sum_{r\geq 0}\beta_r(L_2)L_1^{-r} - 1 + \sum_{n\geq 1}\left(\frac{L_1}{2R_\varepsilon}\right)^{2n-1}\sum_{r\geq 1} D_{n,r}(L_2)L_1^{-r}, \tag{9.10}$$

where every polynomial $D_{n,r}$ is given by the finite extraction

$$\boxed{D_{n,r}(L_2) = [t^r]\,\mathcal{W}(t; L_2)^{2n-1}\sum_{p=1}^{2n-1} e_{n,p}t^p\mathcal{L}(t; L_2)^{-p}.} \tag{9.11}$$

Together with equations (6.7), (9.5) and (9.6), this gives a general formula for every coefficient of every inverse sector.

Proof. The core term follows from $X = (2R_\varepsilon/L_1)\mathcal{W}^{-1}$, and equation (9.6) is precisely reciprocal-series division. For the n-th inverse block,

$$e_n(\Lambda)X^{-(2n-1)} = \left(\frac{L_1}{2R_\varepsilon}\right)^{2n-1}\mathcal{W}^{2n-1}\sum_{p=1}^{2n-1} e_{n,p}L_1^{-p}\mathcal{L}^{-p}.$$

Set $t = L_1^{-1}$ and extract its coefficient. At any fixed r, only finitely many α_j and $e_{n,p}$ are needed. □

Corollary 9.2 (Leading term of each algebraic sector)**.** *The leading term in the n-th inverse-power sector is*

$$-a_n\frac{L_1^{2n-2}}{(2R_\varepsilon)^{2n-1}}. \tag{9.12}$$

Proof. In equation (9.11), the first nonzero power is $r = 1$, arising from $e_{n,1} = -a_n$ and the constant terms of $\mathcal{W}, \mathcal{L}$. □

Summary

The inverse contains two nested grids:

$$\frac{R_\varepsilon}{L_1}\sum_{r\geq 0}\frac{\mathbb{Q}[L_2]}{L_1^r} \quad \text{and} \quad \sum_{n\geq 1}\frac{L_1^{2n-1}}{R_\varepsilon^{2n-1}}\sum_{r\geq 1}\frac{\mathbb{Q}[L_2]}{L_1^r}.$$

Unsigned Stirling numbers generate the first grid; Bernoulli numbers generate the sector labels n; Bell and Lagrange coefficients generate the internal polynomials.

Chapter 10

Beyond all algebraic orders

10.1 Optimal forward remainder

The exact Borel kernel shows that the formal log series has nearest singularities at $\pm\mathrm{i}\pi$. If it is truncated near its least term, $N \approx \pi u/2$, then

$$R_N(u) = \mathrm{O}\left(u^{-1/2}\mathrm{e}^{-\pi u}\right) \qquad (u \to +\infty).$$

A more refined exponentially improved expansion resolves R_N into terminant contributions associated with the pole pairs $\pm\mathrm{i}\pi m$. Such expansions are standard for the gamma function and its reciprocal [9]; the shifted form here has the explicit Borel transform equation (3.1), so the singularity locations and residues are already known.

10.2 Transport through the inverse map

Suppose an algebraically truncated forward phase is written

$$\Phi(u) = \frac{u}{2}(\log u - 1) + A_\varepsilon + \Omega_{\mathrm{alg}}(u) + \mathcal{R}(u),$$

where $\mathcal{R}$ is an exponentially improved remainder. Let u_{alg} solve the equation without $\mathcal{R}$. One Newton linearization gives

$$u - u_{\mathrm{alg}} \sim -\frac{\mathcal{R}(u_{\mathrm{alg}})}{\frac{1}{2}\log u_{\mathrm{alg}} + \Omega'_{\mathrm{alg}}(u_{\mathrm{alg}})} \sim -\frac{2\mathcal{R}(X)}{\Lambda}. \tag{10.1}$$

Thus the optimally truncated inverse remainder on the positive real axis has the scale

$$\mathrm{O}\left(\frac{X^{-1/2}\mathrm{e}^{-\pi X}}{\Lambda}\right). \tag{10.2}$$

For a full terminant transseries, retain the exact remainder in the normalized equation:

$$\Lambda E + H(E) + \sum_{k\geq 1} a_k q^{2k}(1+E)^{-(2k-1)} + 2q\,\mathcal{R}(X(1+E)) = 0. \tag{10.3}$$

Coefficient extraction in the enlarged transmonomial semigroup then proceeds by the same Bell/Faà di Bruno operations as before. At first order, every forward terminant sector is multiplied by $-2/\Lambda$; nonlinear products generate higher sectors.

Interpretive warning

The complex exponentials associated with Borel poles are $\mathrm{e}^{\mp \mathrm{i}\pi m u}$, together with direction-dependent terminants. On the positive real axis it is the terminant-weighted combination, not a bare oscillatory exponential, whose optimally truncated magnitude is $\mathrm{e}^{-\pi u}$. Conflating these two descriptions gives incorrect Stokes statements.

Chapter 11

The inverse of the discrete sequence

11.1 Parity candidates

For $y > 1$, let

$$x_0(y) := D_0^{-1}(y), \qquad x_1(y) := D_1^{-1}(y).$$

The least even and odd indices whose double factorials exceed y are

$$E(y) := 2\left\lceil \frac{x_0(y)}{2} \right\rceil, \tag{11.1}$$

$$O(y) := 2\left\lceil \frac{x_1(y)-1}{2} \right\rceil + 1. \tag{11.2}$$

Theorem 11.1 (Exact generalized inverse). *Let*

$$N(y) := \min\{n \in \mathbb{N} : n!! \geq y\}.$$

For $y > 1$,

$$\boxed{N(y) = \min\{E(y), O(y)\}.} \tag{11.3}$$

The real quantities x_0, x_1 *in equations (11.1) and (11.2) may be evaluated by the transseries equation (6.6).*

Proof. On each parity class, $n!!$ is strictly increasing, so equation (11.1) is the least admissible even index and equation (11.2) the least admissible odd index. The least admissible index without a parity restriction is their minimum. □

11.2 Rounding and certification

An asymptotic approximation can be rounded safely only when its error interval does not cross a parity lattice point. A robust procedure is:

Algorithm 11.2 (Certified discrete inversion).

1. *Compute truncated approximations to* $x_0(y)$ *and* $x_1(y)$.
2. *Form the two neighboring even integers around* x_0 *and the two neighboring odd integers around* x_1.
3. *Evaluate* $\log D_\varepsilon(n)$ *at those candidates, using either* $\log\Gamma$ *or the exact recurrence.*
4. *Select the least candidate satisfying* $\log D_\varepsilon(n) \geq \log y$.

Only a constant number of exact or high-precision checks is required. The asymptotic expansion supplies the location; the final comparisons remove all rounding ambiguity.

Chapter 12

A single analytic interpolation and its oscillatory inverse

12.1 The all-integer interpolation

OEIS records the analytic interpolation

$$\mathscr{D}(x) := 2^{x/2}\left(\frac{2}{\pi}\right)^{\frac{1}{2}\sin^2(\pi x/2)}\Gamma\left(\frac{x}{2}+1\right), \tag{12.1}$$

which agrees with $n!!$ at every nonnegative integer [1]. In terms of $u = x+1$,

$$\log \mathscr{D}(x) = \frac{u}{2}(\log u - 1) + A_0 + \Omega(u) + P(x), \tag{12.2}$$

where

$$P(x) := \frac{1}{4}(1-\cos\pi x)\log\left(\frac{2}{\pi}\right). \tag{12.3}$$

At even integers $P = 0$; at odd integers $P = \frac{1}{2}\log(2/\pi)$, exactly converting A_0 to A_1. Since

$$\frac{d}{dx}\log \mathscr{D}(x) = \frac{1}{2}\left(\log 2 + \psi\left(\frac{x}{2}+1\right)\right) + P'(x) \sim \frac{1}{2}\log x,$$

and P' is bounded, $\mathscr{D}$ is strictly increasing on a real tail. Throughout this chapter, $\mathscr{D}^{-1}$ denotes that tail inverse as $y \to +\infty$, not a globally single-valued inverse near the origin.

Interpretive warning

The periodic phase $P(x) = \mathrm{O}(1)$ displaces the inverse by $\mathrm{O}(\Lambda^{-1})$. The Bernoulli correction displaces it only by $\mathrm{O}((X\Lambda)^{-1})$. Therefore the inverse of the single interpolation $\mathscr{D}$ is *not* obtained by simply forgetting parity in the branchwise formulas.

12.2 Phase-coordinate Lagrange inversion

Let

$$R_0 := \log y - A_0, \qquad h(R) := \frac{2R}{\mathrm{W}_0(2R/\mathrm{e})}.$$

Thus h is the inverse of $u \mapsto \frac{1}{2}u(\log u - 1)$. Define

$$\mathcal{A}(R) := P(h(R)-1) + \Omega(h(R)). \tag{12.4}$$

With $s = \frac{1}{2}u(\log u - 1)$, the interpolation equation becomes

$$R_0 = s + \mathcal{A}(s).$$

Theorem 12.1 (All-orders phase-coordinate reversion)**.** *Introduce a bookkeeping parameter τ in $R_0 = s + \tau\mathcal{A}(s)$. Its formal inverse is*

$$s = R_0 + \sum_{m=1}^{\infty} \frac{(-\tau)^m}{m!} \left(\frac{d}{dR_0}\right)^{m-1} \mathcal{A}(R_0)^m. \tag{12.5}$$

Setting $\tau = 1$ as an asymptotic expansion gives

$$\boxed{\mathscr{D}^{-1}(y) \sim h\left(R_0 + \sum_{m=1}^{\infty} \frac{(-1)^m}{m!} \left(\frac{d}{dR_0}\right)^{m-1} \mathcal{A}(R_0)^m\right) - 1.} \tag{12.6}$$

This formula generates all pure periodic and mixed Bernoulli–periodic coefficients.

Proof. This is the Lagrange reversion formula for the near-identity map $s \mapsto s + \tau\mathcal{A}(s)$, followed by composition with h. The parameter τ makes the formal meaning explicit. In the asymptotic regime, each derivative with respect to R_0 introduces factors controlled by $h'(R_0) = 2/\Lambda$, so the resulting expansion is ordered by inverse powers of Λ, with additional powers of X^{-1} from Ω and from curvature of the core. $\square$

12.3 Leading oscillatory hierarchy

Let $X = h(R_0)$, $\Lambda = \log X$, and abbreviate $P = P(X-1)$, with primes denoting derivatives of P at $X - 1$. The leading pure-periodic displacement is

$$\begin{aligned} \mathscr{D}^{-1}(y) - (X-1) \sim & -\frac{2P}{\Lambda} + \frac{4PP'}{\Lambda^2} \\ & - \frac{8}{\Lambda^3}\left(P(P')^2 + \frac{1}{2}P^2P''\right) + \cdots, \end{aligned} \tag{12.7}$$

up to mixed corrections suppressed by at least one power of X^{-1}. The first Bernoulli contribution is

$$-\frac{2\Omega(X)}{\Lambda} \sim \frac{1}{6\Lambda X}.$$

Formula theorem 12.1 is preferable to hand expansion beyond these first terms.

Chapter 13

Complex inverse sheets

13.1 Logarithmic and Lambert branches

For complex y, choose a logarithmic sheet

$$R_{\varepsilon,\ell} := \operatorname{Log} y + 2\pi \mathrm{i}\ell - A_\varepsilon, \qquad \ell \in \mathbb{Z},$$

and a Lambert branch W_k. Define

$$v_{k,\ell} := W_k\left(\frac{2R_{\varepsilon,\ell}}{\mathrm{e}}\right), \qquad X_{k,\ell} := \frac{2R_{\varepsilon,\ell}}{v_{k,\ell}}, \qquad \Lambda_{k,\ell} := v_{k,\ell} + 1.$$

In any sector where these choices define a simple local inverse and $X_{k,\ell} \to \infty$, the same coefficient functions $d_{2n-1}(\Lambda)$ give

$$x_{\varepsilon;k,\ell}(y) \sim X_{k,\ell} - 1 + \sum_{n\geq 1} d_{2n-1}(\Lambda_{k,\ell}) X_{k,\ell}^{-(2n-1)}.$$

The coefficients are universal; only the sheet data change.

13.2 Branch caveats

The gamma function is meromorphic and $\log \Gamma$ is multivalued. Consequently, not every formal pair (k, ℓ) corresponds to a globally distinct inverse branch throughout the plane. The formula is local and sectorial. Branch collisions occur where $D'_\varepsilon(x) = 0$, equivalently

$$\log 2 + \psi(x/2 + 1) = 0.$$

Near such points the simple inverse expansion must be replaced by a Puiseux-type local analysis.

Chapter 14

Symbolic coefficient algorithms

14.1 Forward coefficients

The forward logarithmic coefficients are direct Bernoulli expressions. The multiplicative coefficients can be generated in $\mathrm{O}(M^2)$ arithmetic operations by equation (4.12).

Algorithm 14.1 (Forward coefficient table)**.** *Given M:*

1. *Compute c_k for $1 \leq k \leq \lceil M/2 \rceil$.*
2. *Set $\gamma_{2k-1} = c_k$, $\gamma_{2k} = 0$.*
3. *Set $s_0 = 1$.*
4. *For $m = 1, \ldots, M$, compute*

$$s_m = \frac{1}{m} \sum_{j=1}^{m} j \gamma_j s_{m-j}.$$

14.2 Inverse coefficients

A Wolfram Language implementation of the triangular recurrence is:

```
ClearAll[c, a, z, lam, ser, e, n, k];

c[k_Integer?Positive] :=
  (1 - 2^(2 k - 1)) BernoulliB[2 k]/(2 k (2 k - 1));

a[k_Integer?Positive] := 2 c[k];

InverseBlocks[N_Integer?Positive] := Module[
  {ser = 0, coeff, en, table = <||>},
  Do[
    coeff = SeriesCoefficient[
      (1 + ser) Log[1 + ser] - ser +
      Sum[a[k] z^k (1 + ser)^(-(2 k - 1)), {k, 1, n}],
      {z, 0, n}
    ];
    en = Together[-coeff/lam];
    AssociateTo[table, n -> en];
    ser = Together[ser + en z^n],
    {n, 1, N}
  ];
```

```
  table
];
```

Here $z = q^2$, ‘lam‘ represents Λ, and the returned value at key n is $e_n = d_{2n-1}$.

14.3 Coefficient-audit identities

Any implementation should verify the following independent checks:

$$e_1 = -a_1/\Lambda, \tag{14.1}$$

$$\lim_{\Lambda\to\infty} \Lambda e_n(\Lambda) = -a_n, \tag{14.2}$$

$$\deg_{\Lambda^{-1}} e_n \le 2n - 1, \tag{14.3}$$

$$\left[q^{2j}\right] \Lambda E_N + H(E_N) + \sum_{k=1}^{N} a_k q^{2k}(1 + E_N)^{-(2k-1)} = 0 \quad (1 \le j \le N). \tag{14.4}$$

The last identity is a residual certificate rather than a comparison against a precomputed table.

Chapter 15

Numerical validation

15.1 Forward logarithmic errors

The following table compares the exact $\log D_\varepsilon(n)$ with the centered logarithmic expansion truncated after N Bernoulli terms. Entries are approximation minus exact value.

Table 15.1: Errors in the centered logarithmic expansion.

ε	n	$N=0$	$N=1$	$N=2$	$N=3$	$N=4$
0	20	3.96616×10^{-3}	-2.09362×10^{-6}	5.98269×10^{-9}	-4.14413×10^{-11}	5.30661×10^{-13}
1	21	3.78606×10^{-3}	-1.82137×10^{-6}	4.74399×10^{-9}	-2.99567×10^{-11}	3.49749×10^{-13}
0	100	8.25064×10^{-4}	-1.88702×10^{-8}	2.34020×10^{-12}	-7.04697×10^{-16}	3.92938×10^{-19}

The alternating signs agree with equation (4.4).

15.2 Inverse errors

For each test, y was generated from an exact branch value $y = D_\varepsilon(x)$. The table gives the error of the inverse expansion through $d_{2N-1}X^{-(2N-1)}$; $N = 0$ means the Lambert core alone.

Table 15.2: Errors of the parity-resolved inverse transseries.

ε	x	$N=0$	$N=1$	$N=2$	$N=3$	$N=4$
0	20	-2.60549×10^{-3}	1.75197×10^{-6}	-4.77514×10^{-9}	3.10287×10^{-11}	-3.81720×10^{-13}
1	21	-2.44974×10^{-3}	1.49560×10^{-6}	-3.71702×10^{-9}	2.20413×10^{-11}	-2.47399×10^{-13}
0	100	-3.57548×10^{-4}	9.58050×10^{-9}	-1.14691×10^{-12}	3.31593×10^{-16}	-1.80329×10^{-19}
1	101	-3.53289×10^{-4}	9.27850×10^{-9}	-1.08916×10^{-12}	3.08781×10^{-16}	-1.64658×10^{-19}

The data validate the predicted odd inverse-power grid and show that the first Bernoulli block removes roughly three to five decimal orders in these examples.

Chapter 16

Generalization to multifactorials

16.1 Residue-class branches

The double-factorial calculation is the $p = 2$ case of a common multifactorial normal form. Let $p \geq 1$, and choose a residue representative $r \in \{1, \ldots, p\}$. On integers $x \equiv r \pmod{p}$, define

$$x!_{(p)} = x(x-p)(x-2p)\cdots r.$$

Its natural residue-class continuation is

$$D_{p,r}(x) := C_{p,r}p^{x/p}\Gamma\left(\frac{x}{p}+1\right), \qquad C_{p,r} := \frac{p^{1-r/p}}{\Gamma(r/p)}. \tag{16.1}$$

All residue branches again differ only by constants.

Set

$$u := x + \frac{p}{2}, \qquad A_{p,r} := \log C_{p,r} + \frac{1}{2}\log\left(\frac{2\pi}{p}\right).$$

Then

$$\log D_{p,r}(x) = \frac{u}{p}(\log u - 1) + A_{p,r} + \Omega_p(u), \tag{16.2}$$

where

$$\Omega_p(u) \sim \sum_{k\geq 1} \frac{c_k^{(p)}}{u^{2k-1}}, \tag{16.3}$$

$$c_k^{(p)} := \frac{p^{2k-1}B_{2k}(\frac{1}{2})}{2k(2k-1)}. \tag{16.4}$$

Its exact Borel kernel is

$$\Omega_p(u) = \int_0^\infty \mathrm{e}^{-us}\,\frac{\frac{ps}{2\sinh(ps/2)} - 1}{ps^2}\,ds. \tag{16.5}$$

The nearest Borel poles lie at $s = \pm 2\pi\mathrm{i}/p$, so the optimal forward remainder scale is $\mathrm{e}^{-2\pi u/p}$.

16.2 Universal multifactorial inverse recurrence

Let

$$R = \log y - A_{p,r}, \qquad v = \mathrm{W}_0\left(\frac{pR}{\mathrm{e}}\right), \qquad X = \frac{pR}{v}, \qquad \Lambda = v + 1.$$

Then X solves $\frac{X}{p}(\log X - 1) = R$. With $u = X(1+E)$ and $q = X^{-1}$, the normalized equation is

$$\Lambda E + H(E) + \sum_{k\geq 1} a_k^{(p)} q^{2k}(1+E)^{-(2k-1)} = 0, \qquad a_k^{(p)} := p\, c_k^{(p)}. \tag{16.6}$$

Every theorem of theorems 6.2 to 6.5 remains valid after replacing a_k by $a_k^{(p)}$. Thus the double-factorial recurrence is one member of a universal centered multifactorial inversion calculus.

Chapter 17

Conclusions

The main structural results can be compressed into five statements.

1. The shift $u = x + 1$ places both parity branches in the same centered gamma normal form. Parity survives only in A_ε.
2. The universal correction has the exact Borel transform

$$\widehat{\Omega}(s) = \frac{s/\sinh s - 1}{2s^2},$$

   hence explicit Bernoulli/eta coefficients, a signed remainder, and a completely known pole lattice.
3. Exact inversion of the dominant phase gives the Lambert core $X = 2R_\varepsilon / \mathrm{W}_0(2R_\varepsilon/\mathrm{e})$. Corrections lie only at odd powers $X^{-1}, X^{-3}, \ldots$.
4. Every inverse coefficient is determined by a triangular Bell recurrence and, independently, by a closed Lagrange–Bürmann extraction. Expanding the Lambert core with Stirling cycle numbers completes the iterated-logarithm transseries.
5. The parity-resolved inverse, the discrete generalized inverse, and the inverse of a single oscillatory interpolation are different objects. Their relation is explicit, so no ambiguity needs to be hidden.

The coefficient formulas are suitable for symbolic generation to arbitrary order, while the residual bounds and exact monotonicity make them suitable for certified numerical inversion. The same centered mechanism extends without essential change to every residue class of every multifactorial.

Appendix A

Coefficient tables

A.1 Logarithmic Bernoulli coefficients

Table A.1: The coefficients c_k and $a_k = 2c_k$.

k	c_k	a_k
1	$-1/12$	$-1/6$
2	$7/360$	$7/180$
3	$-31/1260$	$-31/630$
4	$127/1680$	$127/840$
5	$-511/1188$	$-511/594$
6	$1414477/360360$	$1414477/180180$
7	$-8191/156$	$-8191/78$
8	$118518239/122400$	$118518239/61200$

A.2 Two additional inverse blocks

For reference,

$$\begin{aligned} d_{11}(\Lambda) = -\frac{1}{5884534656000\Lambda^{11}} \Big(& 46195687238400\Lambda^{10} + 8854370366400\Lambda^{9} + 2171643318000\Lambda^{8} \\ & + 605515022112\Lambda^{7} + 212869408752\Lambda^{6} + 84136852800\Lambda^{5} + 35589153600\Lambda^{4} \\ & + 14234019800\Lambda^{3} + 4868113250\Lambda^{2} + 1213962750\Lambda + 165540375 \Big), \end{aligned} \tag{A.1}$$

$$\begin{aligned} d_{13}(\Lambda) = \frac{1}{35307207936000\Lambda^{13}} \Big(& 3707709489792000\Lambda^{12} + 571674673451520\Lambda^{11} \\ & + 109005523604352\Lambda^{10} + 22558151064384\Lambda^{9} + 5893694526096\Lambda^{8} + 1791313043520\Lambda^{7} \\ & + 647003557200\Lambda^{6} + 251996865120\Lambda^{5} + 98865967200\Lambda^{4} + 35397962600\Lambda^{3} \\ & + 10534674150\Lambda^{2} + 2254502250\Lambda + 260134875 \Big). \end{aligned} \tag{A.2}$$

Appendix B

Proof of the phase reversion formula

For completeness, introduce a formal parameter τ and consider

$$Y = s + \tau A(s).$$

Write $s = Y + \sigma$. Then

$$\sigma = -\tau A(Y + \sigma).$$

The Lagrange–Bürmann formula applied to $\sigma = \tau\phi(\sigma)$ with $\phi(\sigma) = -A(Y + \sigma)$ yields

$$\sigma = \sum_{m \geq 1} \frac{\tau^m}{m!} \frac{d^{m-1}}{d\sigma^{m-1}} \big[-A(Y + \sigma)\big]^m \bigg|_{\sigma=0}.$$

Translation invariance of differentiation gives

$$s = Y + \sum_{m \geq 1} \frac{(-\tau)^m}{m!} \left(\frac{d}{dY}\right)^{m-1} A(Y)^m.$$

This proves equation (12.5) and also explains why derivatives, rather than ordinary powers alone, organize perturbations of an invertible phase.

Appendix C

Notation index

Symbol	Meaning
$D_\varepsilon(x)$	Parity-resolved continuation, equation (2.1).
$u = x + 1$	Centered variable for the double factorial.
A_ε	Parity constant: $\frac{1}{2}\log\pi$ or $\frac{1}{2}\log 2$.
$\Omega(u)$	Universal centered gamma correction, equation (2.4).
c_k	Forward logarithmic coefficient, equation (4.2).
s_m	Forward multiplicative coefficient, equation (4.10).
R_ε	Shifted target logarithm $\log y - A_\varepsilon$.
v	Lambert value $\mathrm{W}_0(2R_\varepsilon/\mathrm{e})$.
X	Exact dominant inverse core $2R_\varepsilon/v$.
Λ	Core logarithm $\log X = v + 1$.
q	Small inverse-core parameter X^{-1}.
E	Relative inverse correction $u/X - 1$.
$e_n = d_{2n-1}$	Coefficient of q^{2n} in E, equivalently of $X^{-(2n-1)}$ in the inverse.
L_1, L_2	$\log(2R_\varepsilon/\mathrm{e})$ and $\log L_1$.
$\widehat{B}_{r,m}$	Ordinary partial Bell convolution, equation (6.8).

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